

Full System Reliability Analysis (New Uploaded Labels)

Generated: 2026-02-13 (local run)

1) What was executed

1. Replaced eval labels with uploaded files:

- `evalset/gold.jsonl`
- `evalset/silver.jsonl`

2. Rebuilt index:

- `python -m verified\_agentic\_rag.cli ingest --reset --scope docs`
- Result: 56 chunks indexed

3. Ran evaluation suites:

- Retrieval: `reports/runs/20260213\_uploaded\_labels/suite/20260213\_uploaded\_labels\_suite\_retrieval.json`
- Full: `reports/runs/20260213\_uploaded\_labels/suite/20260213\_uploaded\_labels\_suite\_full.json`

4. Ran tests:

- `pytest -q`
- Result: 39/39 passed

5. Generated artifacts:

- CSV: `reports/runs/20260213\_uploaded\_labels/tables/20260213\_uploaded\_labels\_question\_results.csv`
- Summary report:
`reports/runs/20260213\_uploaded\_labels/analysis/20260213\_uploaded\_labels\_evaluation\_summary.md`
- Summary PDF:
`reports/runs/20260213\_uploaded\_labels/analysis/20260213\_uploaded\_labels\_evaluation\_summary.pdf`
- Full Q&A audit: `reports/runs/20260213\_uploaded\_labels/analysis/20260213\_uploaded\_labels\_answer\_audit.md`
- Full Q&A audit PDF:
`reports/runs/20260213\_uploaded\_labels/analysis/20260213\_uploaded\_labels\_answer\_audit.pdf`
- Figures:
 - `reports/runs/20260213\_uploaded\_labels/figures/20260213\_uploaded\_labels\_pass\_rate\_by\_dataset\_mode.png`
 - `reports/runs/20260213\_uploaded\_labels/figures/20260213\_uploaded\_labels\_behavior\_confusion\_full\_mode.png`
 - `reports/runs/20260213\_uploaded\_labels/figures/20260213\_uploaded\_labels\_rule\_pass\_breakdown\_full\_mode.png`
 - `reports/runs/20260213\_uploaded\_labels/figures/20260213\_uploaded\_labels\_retrieval\_ranking\_metrics\_distribution.png`
 - `reports/runs/20260213\_uploaded\_labels/figures/20260213\_uploaded\_labels\_context\_precision\_distribution.png`

2) Evaluation algorithm used

The evaluator runs two modes over both datasets.

Retrieval mode

- Runs retrieval pipeline without LLM answering.
- Checks:

- evidence recall at k (did we retrieve required evidence labels?)
- ranking quality (MRR, nDCG, rank-aware precision)
- expected value presence in retrieved context (for value-based items)
- `pass\_overall` in retrieval mode requires expected doc/evidence/value rules.

Full mode (strict end-to-end)

- Runs full agent path: retrieve -> decide answer/refuse -> produce citations -> verify grounding.
- Checks:
 - refusal correctness (`pass\_refusal\_rule`)
 - citation rule (`pass\_citation\_rule`)
 - expected value/rule matching (`pass\_expected\_value\_rule`)
 - citation overlap with gold evidence (`pass\_citation\_overlap\_rule`)
 - verification gate (`pass\_verification\_rule`)
- `pass\_overall` requires all strict rules to pass.

3) Dataset size and composition

	Dataset	Total	metadata	procedure	definition	multi_chunk	out_of_scope	other
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	gold	42	19	10	3	0	8	2 (`scope`)
	silver	142	107	14	3	3	15	0

4) Aggregate results

	Dataset	Retrieval pass	Full pass	Retrieval pass rate	Full pass rate
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	gold	38/42	21/42	90.48%	50.00%
	silver	131/142	81/142	92.25%	57.04%

Retrieval metrics (aggregate):

- Gold retrieval: Recall@k 1.0000, MRR 0.9706, nDCG@k 0.9783, ranked precision 0.9706
- Silver retrieval: Recall@k 1.0000, MRR 0.9961, nDCG@k 0.9971, ranked precision 0.9961

Full-mode evidence metrics (aggregate):

- Gold full: Recall@k 1.0000, MRR 0.9853, nDCG@k 0.9891, ranked precision 0.9853
- Silver full: Recall@k 1.0000, MRR 1.0000, nDCG@k 1.0000, ranked precision 1.0000

Interpretation:

- Retrieval quality is very high.
- End-to-end pass drops because strict answer/citation/refusal rules fail on many questions, not because evidence is missing.

5) Behavior and reliability findings

What is strong

- Retriever consistently finds evidence (Recall@k = 1.0 on both sets).
- Out-of-scope behavior is mostly good:
 - gold: 7/8 passed
 - silver: 15/15 passed
- Verification failures are zero in this run.

Main failure categories (full mode)

- Gold failures (21 total):
 - `pass\_expected\_value\_rule`: 20
 - `pass\_refusal\_rule`: 16
 - `pass\_citation\_overlap\_rule`: 15
- Silver failures (61 total):
 - `pass\_expected\_value\_rule`: 61
 - `pass\_citation\_overlap\_rule`: 42
 - `pass\_refusal\_rule`: 34

Failure pattern analysis

1. False refusals on answerable questions (largest issue)

- Gold: 15 answer-expected questions were refused.
- Silver: 34 answer-expected questions were refused.
- Many of these had perfect retrieval evidence metrics, so failure is in answering/gating stage.

2. Value normalization or label mismatch

- Several expected values are abbreviations or formats not matching corpus phrasing.
- Examples:

- expected `P&F` vs answer `PERSONNEL`
- expected `03/2024` vs answer `11.03.2025`

- In retrieval-mode failures, `corpus\_contains\_expected\_value` is often false, indicating label-text mismatch rather than retrieval miss.

3. Cross-policy citation leakage on some metadata prompts

- Example: Provision IT committee question answered with Data Protection citation/answer (`PERSONNEL`) instead of target policy evidence label.
- This causes citation-overlap failure even when text appears plausible.

4. Rare false positive on refusal

- 1 gold out-of-scope question was incorrectly answered:
 - Q: "What is the latest version of the OpenAI API?"
 - A: "The version is 2025.[docs/txt/Equality_Diversity.txt:1-100]"

6) Example exact Q&A (from full-mode output)

Correct examples

- `g001`
 - Q: What is the policy group for the Data Protection - Employees policy?
 - A: The policy group is Employees.[docs/txt/Data_Protection_-_Employees.txt:1-100]
- `g004`
 - Q: What is the next review date guidance for the Data Protection - Employees policy?
 - A: The review frequency is Annual or if required by legislation.[docs/txt/Data_Protection_-_Employees.txt:1-100]

Incorrect examples

- `g003` (value mismatch)
 - Expected: 03/2024
 - A: The last updated is 11.03.2025.[docs/txt/Data_Protection_-_Employees.txt:1-100]
- `g019` (false refusal)
 - Q: Name two lawful bases for processing personal information listed in the Data Protection - Employees policy.
 - A: I cannot answer from the repository based on the retrieved sources.
- `s048` (wrong evidence source)
 - Q: Which committee is responsible for Provision of IT Acceptable Use Policy Employees Members?
 - A: The responsible committee is PERSONNEL.[docs/txt/Data_Protection_-_Employees.txt:1-100]
 - Gold evidence target: Provision IT policy header

7) Critical accuracy conclusion

- Retrieval subsystem reliability: high.
- End-to-end strict reliability on this expanded benchmark: moderate (50-57%), not production-ready yet.
- The current agent is strongest on deterministic metadata and refusal safety.
- The weakest area is converting available evidence into accepted full-mode answers for procedure/definition/narrative items.

8) What has been achieved

You now have:

- much larger and stricter benchmark coverage than before,
- complete reproducible evaluation artifacts (JSON + CSV + figures + PDF),
- a full question-by-question audit including exact agent answers and citations,
- clear diagnostics separating retrieval quality from full-mode answer reliability.

This is a strong engineering milestone because failures are now measurable and classifiable.

9) Immediate next fixes to increase full-mode pass rate

1. Reduce false refusals on answerable narrative/procedure questions (tune retrieval grading/gating thresholds and narrative answer path).
2. Add normalization mappings in eval for known canonical aliases (for example `P&F` <-> `PERSONNEL`, date formats).
3. Strengthen policy-specific routing so metadata extraction/citations stay in the requested document.

4. Add extra gold narrative items with `required\_phrases` to avoid brittle literal expected strings.